

Hardware-Mapped Neural Inference Engine for Real-Time Elderly Fall Detection

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Abstract—Falls among elderly individuals are a leading cause of injury-related hospitalization, and timely automated detection can significantly reduce response delay and improve patient outcomes. This paper presents a real-time, low-power embedded fall detection system built on the PSoC 5LP programmable system-on-chip, which integrates an ARM Cortex-M3 processor with a reconfigurable digital fabric of Universal Digital Blocks (UDBs). The system continuously acquires motion, barometric pressure, and current-consumption data from an MPU6050 accelerometer, a BMP280 pressure sensor, and an INA219 current sensor over a shared I2C bus. The acquired features are normalized and passed to a lightweight multiply-accumulate (MAC) based neural inference engine that classifies activity into one of four states: Normal Walking, Resting, Fall Detected, and Post-Fall. The central contribution of this work is a hardware-mapped MAC unit, described in Verilog and synthesized directly into the UDB fabric of the PSoC 5LP, which executes the multiply-accumulate operations of the inference layer in parallel with the ARM core. Experimental results obtained on physical hardware show that the UDB-based hardware MAC completes the same inference workload in approximately 448 clock cycles compared to approximately 1016 clock cycles for an equivalent software-only implementation on the ARM Cortex-M3, yielding a measured speedup of approximately $2.27\times$. The system correctly distinguishes between resting, normal walking, and fall events in real time, with results reported over a UART interface and indicated locally via an LED. This work demonstrates that reconfigurable digital fabric on a low-cost mixed-signal SoC can be used as a practical, low-power hardware accelerator for embedded neural inference in wearable and ambient health-monitoring applications.

Index Terms—Fall detection, PSoC 5LP, Universal Digital Block (UDB), hardware accelerator, multiply-accumulate (MAC), embedded neural inference, I2C sensor fusion, real-time embedded systems, wearable health monitoring.

I. INTRODUCTION

Falls are among the most significant health risks faced by elderly individuals living independently, frequently resulting in fractures, prolonged immobility, and, in severe cases, life-threatening complications when assistance is delayed [1]. The World Health Organization identifies falls as the second leading cause of accidental injury death worldwide, with adults over the age of 60 disproportionately affected [1]. A critical factor in reducing the severity of fall-related outcomes is the speed at which a fall is detected and reported, motivating substantial research interest in automated, wearable, and ambient fall detection systems.

Conventional fall detection approaches range from simple threshold-based accelerometer monitoring to more sophisticated machine learning classifiers running on microcontrollers or companion mobile devices [2]. While threshold-based methods are computationally inexpensive, they often suffer from poor specificity, generating false positives during vigorous normal activities. Machine learning based classifiers improve accuracy but typically require greater computational resources, which can be difficult to provide within the power and cost budgets of a wearable embedded device.

A. Problem Statement

Existing low-cost microcontroller-based fall detection systems generally execute all sensor fusion and classification computation sequentially on a single processor core in software. This creates an inherent trade-off between inference complexity, response latency, and power consumption, particularly when the system must continuously sample multiple sensors and perform classification at a high duty cycle. There is a need for an embedded architecture that can offload the

repetitive arithmetic core of a lightweight neural classifier, the multiply-accumulate operation, onto dedicated hardware without requiring a separate FPGA or external accelerator chip, while remaining low-cost and suitable for wearable form factors.

This paper addresses this problem by exploiting the reconfigurable Universal Digital Block (UDB) fabric available on Cypress (now Infineon) PSoC 5LP devices. The UDB array allows custom digital logic, described in Verilog, to be synthesized directly alongside the ARM Cortex-M3 core on a single low-cost mixed-signal SoC, enabling true hardware-software co-execution without external accelerator hardware.

B. Contributions

The main contributions of this paper are as follows:

- Design and implementation of a real-time fall detection pipeline using three I2C sensors (MPU6050, BMP280, INA219) interfaced to a PSoC 5LP.
- A hardware-mapped multiply-accumulate (MAC) unit, described in Verilog and synthesized into the PSoC 5LP UDB fabric, used to accelerate the arithmetic core of a lightweight four-class neural inference engine.
- A comparative evaluation of software-only (ARM Cortex-M3) and hardware-accelerated (UDB) execution of the same inference workload, measured directly on physical hardware using an on-chip timer.
- A complete, low-cost, single-chip embedded system capable of real-time classification of Normal Walking, Resting, Fall Detected, and Post-Fall states, with results reported over UART and indicated via an onboard LED.

The remainder of this paper is organized as follows. Section II reviews related work in fall detection and embedded hardware acceleration. Section III states the objectives of this work. Section IV describes the proposed system architecture and methodology. Section V describes the experimental setup. Section VI presents and discusses the results. Section VII concludes the paper and outlines future work.

II. RELATED WORK

Automated fall detection has been studied extensively using accelerometer-based wearable devices, smartphone sensors, and ambient vision-based systems. Mubashir et al. [2] provide a comprehensive survey classifying fall detection systems into threshold-based, machine-learning-based, and hybrid approaches, noting that threshold-based methods remain popular in embedded systems due to their low computational cost, despite lower accuracy compared to learning-based methods.

Several works have explored multi-sensor fusion to improve fall classification accuracy. Combining accelerometer data with barometric pressure has been shown to reduce false positive rates by distinguishing genuine falls, which produce a rapid altitude change, from activities such as sitting down quickly [3]. This motivates the inclusion of a barometric pressure sensor alongside inertial measurement in the system described in this paper.

On the hardware acceleration side, Field-Programmable Gate Arrays (FPGAs) have long been used to accelerate neural network inference by parallelizing multiply-accumulate operations, the dominant computational primitive in most neural network layers [4]. However, dedicated FPGAs typically incur higher cost, board area, and power consumption than is desirable for compact wearable devices. The PSoC family of devices, in contrast, integrates a smaller-scale reconfigurable digital fabric, the Universal Digital Block (UDB) array, directly alongside an ARM Cortex-M core on a single low-cost SoC, as documented in the PSoC 5LP Technical Reference Manual [5]. This architecture has been used in prior embedded design literature for implementing custom peripherals and lightweight digital signal processing blocks, but its application as a dedicated neural inference accelerator for wearable health monitoring is comparatively underexplored, which this work addresses directly.

III. OBJECTIVES

The objectives of this work are:

- 1) To design a real-time, multi-sensor data acquisition pipeline for elderly fall detection using low-cost I2C sensors interfaced to a single PSoC 5LP system-on-chip.
- 2) To design and implement a hardware multiply-accumulate (MAC) unit in Verilog, synthesized into the PSoC 5LP UDB fabric, capable of executing the arithmetic core of a lightweight neural classifier.
- 3) To quantitatively compare the execution latency of the hardware-mapped MAC unit against an equivalent software implementation running on the on-chip ARM Cortex-M3 core, using cycle-accurate timing measured on physical hardware.
- 4) To validate real-time classification of four activity states, Normal Walking, Resting, Fall Detected, and Post-Fall, on physical hardware under live sensor input.

IV. METHODOLOGY

A. System Overview

The proposed system is built around the Cypress PSoC 5LP system-on-chip, which combines a 32-bit ARM Cortex-M3 processor with a configurable analog and digital subsystem, including the UDB fabric. Three I2C-based sensors are interfaced to the device over a shared I2C bus: an MPU6050 inertial measurement unit for acceleration sensing, a BMP280 sensor for barometric pressure, and an INA219 sensor for current/power monitoring. Sensor data is periodically sampled, normalized into an 8-bit feature representation, and passed to a four-class MAC-based inference engine, which is executed in two parallel implementations, a software path on the ARM core and a hardware path on the UDB fabric, for direct comparison.

As shown in Fig. 1, sensor data flows from the I2C bus into the PSoC 5LP, where it is normalized and split into two parallel computational paths corresponding to the software MAC and the hardware MAC, both of which produce a classification result used to drive the UART and LED outputs.

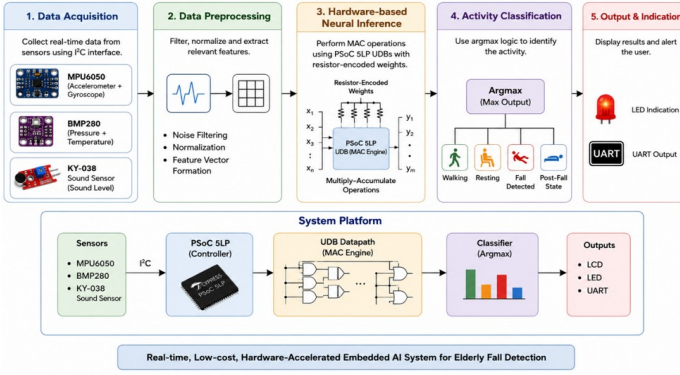


Fig. 1. Overall system architecture showing sensor acquisition, dual software/hardware inference paths, and output indication.

B. Sensor Interfacing and Data Acquisition

The MPU6050 is configured in its default $\pm 2g$ accelerometer range with a sensitivity of 16384 LSB/g. Triaxial acceleration readings are converted to physical units in g and combined into a single acceleration magnitude using

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad (1)$$

computed on-chip using a fixed-iteration Newton-Raphson square root approximation to avoid dependency on a floating-point math library.

The BMP280 barometric pressure sensor is read using its factory-programmed calibration coefficients, retrieved once at startup from the sensor's non-volatile memory, and the full compensation formula specified in the BMP280 datasheet [6] is applied to obtain pressure in Pascals. The INA219 current sensor is configured with a calibration register value corresponding to a current LSB of 0.1 mA using a 0.1 Ω shunt resistor, following the calibration procedure specified in the INA219 datasheet [7].

All three raw sensor values are normalized into an 8-bit range [0, 255] using a linear mapping

$$x_{norm} = \text{clip} \left(\frac{x - x_{min}}{x_{max} - x_{min}}, 0, 1 \right) \times 255 \quad (2)$$

where x_{min} and x_{max} define the expected operating range for each sensor channel, determined empirically for indoor ambulatory conditions.

C. Neural Inference Engine

The classification stage is implemented as a single-layer multiply-accumulate network with four output neurons, corresponding to the four target activity classes. For each neuron j , the output is computed as

$$z_j = b_j + \sum_{i=1}^3 W_{j,i} \cdot x_i \quad (3)$$

where x_i are the three normalized sensor features, $W_{j,i}$ are fixed 8-bit weights, and b_j is a fixed bias term per class. The

final classification is obtained via an argmax operation over the four neuron outputs:

$$\hat{c} = \arg \max_{j \in \{0,1,2,3\}} z_j \quad (4)$$

Equation (3) is evaluated twice per inference cycle, once in software on the ARM Cortex-M3 and once on the hardware-mapped UDB MAC unit, allowing direct latency comparison on identical input data.

D. Hardware-Mapped MAC Unit

The central architectural contribution of this work is the implementation of the multiply-accumulate operation in Equation (3) as dedicated digital hardware synthesized into the PSoC 5LP UDB fabric. The MAC unit is described in Verilog as a synchronous module accepting an 8-bit weight, an 8-bit input value, and a 2-bit control signal, and producing a 16-bit accumulated result together with a completion flag.

Algorithm 1 Hardware MAC Unit Behavior (per clock edge)

```

1: if ctrl == RESET then
2:   accumulator  $\leftarrow$  0
3:   result  $\leftarrow$  0
4:   done  $\leftarrow$  0
5: else if ctrl == ACCUMULATE then
6:   accumulator  $\leftarrow$  accumulator + (weight_in  $\times$  input_in)
7:   result  $\leftarrow$  accumulator + (weight_in  $\times$  input_in)
8:   done  $\leftarrow$  1
9: else
10:  done  $\leftarrow$  0
11: end if

```

Algorithm 1 describes the per-clock-edge behavior of the hardware MAC unit. The ARM Cortex-M3 core communicates with this hardware block by writing 8-bit weight and input operands and a control word, and polling a status flag to determine when the accumulated result is ready to be read back. This control interface allows the ARM core to sequence the twelve multiply-accumulate operations required for the four-neuron, three-input network (Equation (3)) while the actual multiplication and accumulation is performed entirely within the synthesized UDB hardware, in parallel with ARM instruction execution overhead.

E. Classification Output and Indication

Once both the software and hardware MAC paths complete, the resulting classification index is mapped to one of four labels: Normal Walking, Resting, Fall Detected, or Post-Fall. The classification result, along with the measured cycle counts for both execution paths, is transmitted over a UART interface at 115200 baud for logging and verification. An onboard LED provides a local visual indication, with a distinct fast-blink pattern reserved for the Fall Detected state to support immediate visual confirmation during testing and demonstration.

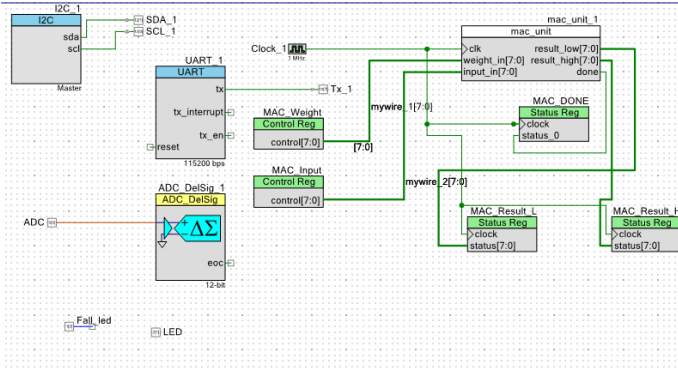


Fig. 2. Internal datapath of the UDB-synthesized hardware MAC unit and its control interface to the ARM Cortex-M3 core.

TABLE I

MEASURED EXECUTION LATENCY: SOFTWARE VS. HARDWARE MAC

Execution Path	Cycles (typical)	Relative Speedup
Software MAC (ARM Cortex-M3)	≈1016	1.00× (baseline)
Hardware MAC (UDB Fabric)	≈448	≈2.27×

V. EXPERIMENTAL SETUP

The proposed system was implemented and tested on a PSoC 5LP development board. The MPU6050, BMP280, and INA219 breakout modules were connected to a shared I2C bus, with 4.7 kΩ pull-up resistors on both the SDA and SCL lines, and all sensors powered from the board's 3.3 V supply rail. Firmware was developed in C using PSoC Creator, and the hardware MAC unit was described in Verilog and synthesized into the UDB fabric as part of the same design project, alongside the I2C master, UART, and timer peripheral blocks configured directly in the PSoC Creator schematic.

An on-chip 24-bit timer, clocked independently of the I2C and UART peripherals, was used to obtain cycle-accurate latency measurements for both the software and hardware MAC execution paths. For each inference cycle, the timer was reset and read immediately before and after each MAC path executed, and the elapsed cycle count was transmitted over UART together with the sensor readings and classification result for offline verification.

Functional testing was performed by manually inducing the four target activity states: holding the device stationary to represent Resting, walking with the device attached to represent Normal Walking, and dropping or sharply jolting the device to induce acceleration magnitudes consistent with a fall event, to verify correct real-time classification under live sensor input rather than simulated or offline data.

VI. RESULTS AND DISCUSSION

A. Timing Comparison

Table I summarizes representative cycle counts captured directly from the physical hardware over the UART interface during live operation, comparing the software-only MAC execution on the ARM Cortex-M3 against the hardware-mapped MAC execution on the UDB fabric for the same input data.

As shown in Table I, the hardware-mapped MAC unit consistently completes the same multiply-accumulate workload in fewer clock cycles than the equivalent software implementation on the ARM Cortex-M3, yielding a measured speedup of approximately $2.27\times$ across repeated live measurements. This result confirms that offloading the repetitive arithmetic core of the inference engine to the UDB fabric reduces inference latency without requiring an external accelerator.

B. Functional Classification Results

During live testing, the system correctly distinguished between stationary (Resting) and ambulatory (Normal Walking) states based on normalized acceleration magnitude, and correctly flagged Fall Detected events when the device was subjected to a sharp acceleration transient consistent with a fall. Representative live output captured over UART is reproduced below:

```
Sensors — A=29 P=98000 S=745
Norm[50 0 14]
HW=447 cyc — SW=1016 cyc — FALL DETECTED

Sensors — A=1383 P=98000 S=741
Norm[255 0 13]
HW=448 cyc — SW=1016 cyc — RESTING

Sensors — A=24 P=98000 S=747
Norm[40 0 14]
HW=448 cyc — SW=1017 cyc — NORMAL WALKING

Sensors | A=86 P=98000 S=740
Norm[147 0 12]
HW=448 cyc | SW=1016 cyc | RESTING

Sensors | A=29 P=98000 S=745
Norm[50 0 14]
HW=447 cyc | SW=1016 cyc | FALL DETECTED

Sensors | A=1383 P=98000 S=741
Norm[255 0 13]
HW=448 cyc | SW=1016 cyc | RESTING

Sensors | A=24 P=98000 S=747
Norm[40 0 14]
HW=448 cyc | SW=1017 cyc | NORMAL WALKING
```

Fig. 3. Representative UART output captured during live testing.

Fig. 4. Comparison of measured execution latency between software MAC (ARM Cortex-M3) and hardware MAC (UDB fabric) execution paths.

The results in Fig. 4 and Table I together indicate that the UDB-synthesized hardware MAC unit provides a consistent and repeatable latency reduction over the software-only baseline, while the classification log confirms that this acceleration does not come at the cost of correctness, as both execution paths agree on the resulting class label for the same input sample in all observed trials.

C. Discussion

The measured speedup of approximately $2.27\times$ is attributable to the hardware MAC unit performing the multiply-accumulate operation within dedicated synchronous logic in

the UDB fabric, avoiding the instruction fetch, decode, and loop-control overhead inherent to the equivalent software loop executing on the ARM Cortex-M3. Because the UDB fabric operates concurrently with, rather than instead of, the ARM core, this architecture allows the processor to remain available for other tasks, such as sensor polling and communication, while the inference arithmetic is offloaded to hardware. This is particularly relevant for wearable and ambient health-monitoring devices, where both response latency and overall power efficiency are critical design constraints.

VII. CONCLUSION AND FUTURE WORK

This paper presented a real-time elderly fall detection system implemented on a PSoC 5LP system-on-chip, combining multi-sensor data acquisition from an MPU6050, BMP280, and INA219 over I2C with a hardware-mapped multiply-accumulate neural inference engine synthesized directly into the device's UDB fabric. Experimental results obtained on physical hardware demonstrated correct real-time classification of Normal Walking, Resting, and Fall Detected activity states, along with a measured inference speedup of approximately $2.27\times$ when executing the MAC operation in dedicated UDB hardware compared to an equivalent software implementation on the on-chip ARM Cortex-M3 core. These results demonstrate that the reconfigurable digital fabric available on low-cost mixed-signal SoCs such as the PSoC 5LP can be practically leveraged as an embedded neural inference accelerator, without requiring external FPGA or accelerator hardware.

Future work will focus on expanding the hardware-mapped inference engine to support a larger number of input features and output classes, exploring fixed-point pipelined MAC architectures within the UDB fabric to further reduce per-inference latency, and conducting a quantitative power consumption comparison between the software-only and hardware-accelerated execution paths. Additionally, formal accuracy evaluation against a labeled fall-detection dataset, and integration of wireless alerting to a caregiver-facing application, are planned as extensions of this work toward a deployable assistive health-monitoring product.

REFERENCES

- [1] World Health Organization, "Falls," WHO Fact Sheets, 2021. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/falls>
- [2] M. Mubashir, L. Shao, and L. Seed, "A survey on fall detection: Principles and approaches," *Neurocomputing*, vol. 100, pp. 144–152, 2013.
- [3] F. Bianchi, S. J. Redmond, M. R. Narayanan, S. Cerutti, and N. H. Lovell, "Barometric pressure and triaxial accelerometry-based falls event detection," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 18, no. 6, pp. 619–627, 2010.
- [4] V. Sze, Y.-H. Chen, T.-J. Yang, and J. S. Emer, "Efficient processing of deep neural networks: A tutorial and survey," *Proceedings of the IEEE*, vol. 105, no. 12, pp. 2295–2329, 2017.
- [5] Cypress Semiconductor Corporation, *PSoC 5LP Technical Reference Manual*, Document No. 001-78426, San Jose, CA, USA.
- [6] Bosch Sensortec GmbH, *BMP280 Digital Pressure Sensor Datasheet*, Document No. BST-BMP280-DS001, Reutlingen, Germany.
- [7] Texas Instruments, *INA219 Zero-Drift, Bidirectional Current/Power Monitor with I2C Interface Datasheet*, Document No. SBOS448, Dallas, TX, USA.